Branch Coverage Report

Field	Detail
Project	Vehicle Parking System (VPS) v1.1 — Group 6
Generated	07 Nov 2026, 17:21 IST
Environment	Python 3.13.18 · Linux x86_64
Source revision	2f4b768
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1. Summary

Metric	Value
Branch points (decision outcomes)	384
Branches taken	382
Branches never taken	2
Partially covered decisions	2
Branch coverage	**99.48 %**

A *branch* is one outcome of a decision (`if` , `while` , `for` , `try/except` , boolean short-circuit) as measured by coverage.py's arc analysis. A decision is *partial* when at least one of its outcomes was never executed.

2. Branch coverage by module

Module	Owner	Branches	Taken	Partial decisions	Branch coverage
`vps/__init__.py`	Angelo	0	0	0	100.0 %
`vps/alerts.py`	Shweta	4	4	0	100.0 %
`vps/allocation.py`	Shweta	26	26	0	100.0 %
`vps/api/__init__.py`	Angelo	0	0	0	100.0 %
`vps/api/admin.py`	Himansh	6	6	0	100.0 %
`vps/api/auth.py`	Angelo	0	0	0	100.0 %
`vps/api/deps.py`	Angelo	0	0	0	100.0 %
`vps/api/edge.py`	Shweta	0	0	0	100.0 %
`vps/api/egress.py`	Akarsh	2	2	0	100.0 %
`vps/api/ingress.py`	Angelo	2	2	0	100.0 %
`vps/api/receipts.py`	Akarsh	0	0	0	100.0 %
`vps/api/reports.py`	Himansh	2	2	0	100.0 %
`vps/api/reservations.py`	Himansh	0	0	0	100.0 %
`vps/api/slots.py`	Shweta	0	0	0	100.0 %

Module	Owner	Branches	Taken	Partial decisions	Branch coverage
`vps/audit.py`	Angelo	8	8	0	100.0 %
`vps/auth.py`	Angelo	34	34	0	100.0 %
`vps/config.py`	Angelo	4	4	0	100.0 %
`vps/context.py`	Angelo	2	2	0	100.0 %
`vps/db.py`	Angelo	4	4	0	100.0 %
`vps/edge_queue.py`	Shweta	8	8	0	100.0 %
`vps/egress.py`	Akarsh	28	28	0	100.0 %
`vps/errors.py`	Angelo	2	2	0	100.0 %
`vps/hardware.py`	Shweta	22	22	0	100.0 %
`vps/ingress.py`	Angelo	16	16	0	100.0 %
`vps/main.py`	Angelo	28	26	2	92.9 %
`vps/payment.py`	Akarsh	26	26	0	100.0 %
`vps/plates.py`	Shweta	4	4	0	100.0 %
`vps/receipt.py`	Akarsh	16	16	0	100.0 %
`vps/reports.py`	Himansh	34	34	0	100.0 %
`vps/reservation.py`	Himansh	34	34	0	100.0 %
`vps/scheduler.py`	Himansh	8	8	0	100.0 %
`vps/security.py`	Angelo	10	10	0	100.0 %
`vps/sensors.py`	Shweta	14	14	0	100.0 %
`vps/tariff.py`	Akarsh	38	38	0	100.0 %
`vps/tls.py`	Angelo	2	2	0	100.0 %
Total		**384**	**382**	**2**	**99.48 %**

3. Branches not taken

Module	From line → to line	Meaning (source of the decision)
`vps/main.py`	84 → 80	jump to line 80 never taken: `if hasattr(mod, "router"):`
`vps/main.py`	128 → 130	jump to line 130 never taken: `if WEB_DIR.exists():`

4. Branch-focused test design

Boundary and decision tests written specifically to drive both outcomes of key decisions:

Decision	Tests driving each outcome
Grace period `minutes <= 15`	0, 1, 15 → free; 16 → charged (`test_grace_period_boundary`)
Daily cap `min(tiered, cap)` and 24 h blocks	3 h (under cap), 10 h/15 h (at cap), 26 h (1 day + 2 h)
Surge window `start <= hour < end`	exit 18:00 IST (in), 11:30 IST (out), rule disabled
Coupon kind / expiry / max uses	PERCENT, FLAT capped at gross, expired, exhausted, unknown

Decision	Tests driving each outcome
Reservation window	47.9 h accepted, 48.1 h and past rejected; check-in early / on time / late
No-show cut-off	start + 29 min (kept) vs + 31 min (released)
Barrier lower vs. loop occupied	occupied → halted; clear → auto-lower
Webhook	valid, bad signature, missing signature, amount mismatch, re-delivery, payment.failed
Edge queue	online forward, offline queue, flush while offline (no-op), 71 h vs 73 h budget
TLS requirement	https scheme, X-Forwarded-Proto https, plain HTTP → 426, /health exempt

5. Artefacts

- Annotated source with partial branches highlighted: `docs/reports/html/coverage/index.html`